

CDFD Runtime Paper 06: CLI-First Platform and Web Architecture

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Abstract

Executive synthesis. The platform direction is CLI first, web app second. The CLI is for serious engine work: reproducible simulations, paper diagnostics, CDFL language workbench commands, debugging, automation, and saved results. An optional web studio (**webapp/**) is a visual layer over the same backend, not a separate physics system. Optional LLM provider calls are interpretive tools above saved deterministic outputs. This paper defines those boundaries and explains how CLI outputs flow into optional visual interfaces.

Greek-symbol convention. Unless a manuscript states a tighter equation-local meaning, Greek symbols are local CDFD variables or coefficients, not universal constants. Here Φ denotes driving flux, Ψ_s denotes the adaptive operating ratio, Ω denotes a domain or state space, and Λ denotes the Life Number where used. The coefficients α , β , and γ denote accumulation or reinforcement, relaxation or decay, and diffusion, coupling, or re-distribution. The symbols δ and ϵ denote perturbation or tolerance scales, while η , λ , μ , ν , ρ , σ , τ , and ϕ denote local efficiency, rate, baseline, frequency, density or correlation, noise or variance, time-scale, and phase or field terms. Subscripts restrict any coefficient to the named pathway, tissue, domain, or experiment.

Runtime evidence path

Prior CDFD mathematics $\longrightarrow$ CDFL/runtime state $\longrightarrow$ bounded output $\longrightarrow$ falsification target.

Figure 1: Conceptual schematic for the runtime papers. The engine translates the mathematical frame from the prior CDFD papers into executable CDFL/runtime diagnostics; candidate outputs remain tied to explicit tests before any stronger claim is made.

1 Prior CDFD Basis

This manuscript does not rederive the mathematical core of CDFD. It uses the Mujjabi Adaptive Operating Ratio and the constrained-vacuum notation introduced in Part I [1],

the torus-knot hierarchy and boundary-mode work from the Part I sequence [2, 3], the public balance law developed in the Part I capstone [4], the Part I transport tests [5], and the Life Number and tri-regime biology bridge from Part II [6]. The runtime papers therefore act as an execution layer: they keep the mathematics connected to commands, outputs, falsification tests, and domain-specific limits.

2 Ordering Principle

The platform order is:

1. core engine;
2. CLI;
3. API or service wrappers;
4. web app.

The current repository has the engine and CLI as primary surfaces. The Streamlit studio under `webapp/` (CDFD Runtime Studio) is optional: it is a visual layer over `runtime/runner.py`, the same kernel, `runtime/diagnostics.py`, CDFL tooling, and domain adapters, and it does not duplicate physics logic.

3 CLI Responsibilities

The CLI supports:

- listing registered domains (`cdfd.py domains`);
- Part II diagnostics export (`cdfd.py diagnostics`);
- running domain demos with JSON payloads or OOL source scenarios;
- validating, executing, linting, formatting, and inspecting CDFL models through `cdfd.py cdf1 ...`;
- exporting JSON result envelopes;
- doctor, gallery, compare, report, and explain commands for public smoke tests and reproducible run artifacts;
- optional provider inventory, status, dry-run prompt audit, and research interpretation through `cdfd.py llm ...`;
- preserving finite-value and provenance metadata on every command.

This is the credibility layer. It gives papers exact commands instead of vague claims that “the engine was run.”

4 CDFD Runtime Studio

The public Streamlit studio (`python -m webapp.run_server`) already provides an initial visual environment: regime-banded trajectories, field heatmaps, Part II source-mix panels, a Life Number calculator, Granger-style causal graphs from run history, the `origins_of_life` domain explorer, and a CDFL Workbench. The workbench lets a user edit a CDFL model, validate it, lint it, run it, format it, inspect its AST, and export JSON/Markdown/HTML result envelopes through the same backend used by the CLI. Future CDFD Studio releases may add richer model building, coherence monitoring, and deeper result browsing on top of the same runtime backend.

The first public demos remain non-medical and visual:

- heat diffusion and thermal flow;
- traffic or infrastructure routing;
- resource-flow networks;
- predator-prey/ecology models;
- photonic or adaptive-surface demos.

Medical applications require stricter validation, privacy, and user-safety boundaries and are not the first public web surface.

5 API Boundary

Any API or service wrapper exposes command-like operations:

- validate a CDFL model;
- run a CDFL model;
- lint, format, and inspect CDFL AST summaries;
- retrieve saved result envelopes;
- inspect finite audits;
- list experiments and selected outputs;
- export reports.

Each endpoint calls `runtime/runner.py` or a later service wrapper around it. The service layer may add authentication, queues, and storage, but not a second physics implementation.

6 API Keys and the AI Boundary

Provider keys belong at the application/provider boundary, not inside the physics kernel. The runtime exposes optional `cdfd.py llm` commands above saved deterministic outputs: provider inventory, provider-key status, dry-run prompt audit, and provider-backed research interpretation. Supported provider shapes are explicit rather than universal. Raw keys are never printed, and `cdfd.py auth` remains only a compatibility alias for provider-key status.

This split keeps the engine clean: `engine/`, CDFL execution, and domain adapters produce deterministic result envelopes; `runtime/llm.py`, the CLI, web layer, VOS, or other governed products may interpret saved outputs through provider adapters. LLM commentary is therefore a separate interpretive artifact, not an input to the engine equations.

7 Vacuum OS

Vacuum OS (VOS) is the proposed operating system layer above CDFD Runtime. VOS is not a second engine. It is the orchestration shell for projects that need apps, CDFL packages, user spaces, experiment queues, provenance stores, and controlled AI assistance. In this architecture, CDFD Runtime remains the mathematical kernel; VOS becomes the operating environment that launches models, manages keys, schedules runs, stores outputs, and connects approved applications to the same CDFD command backend.

8 Experiment Visualization

The experiment outputs already define future panels:

- vacuum hysteresis: M_s residual heatmap and recovery curve;
- knot stability: stability-index trace toward the χ^* attractor;
- OOL phase: $\Lambda = 1$ phase boundary;
- frontier sweep: pattern distribution and candidate triage table;
- gigamarathon sweep: domain coverage dashboard.

The public Streamlit dashboard under `webapp/dashboard.py` implements an initial subset: regime-banded Ψ_s trajectories, Φ and C field heatmaps, Part II aromatic source-mix bars, a Life Number calculator, Granger-style causal edges from run history, the `origins_of_life` domain explorer, and the CDFL Workbench. The web app presents these as model diagnostics and falsification targets, not as empirical proof.

9 Conclusion

The platform makes one thing unmistakable: CDFD is executable. The CLI establishes that discipline. The optional web studio makes the same discipline visible without weakening the runtime boundary or replacing command-line reproducibility.

References

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